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Stress–Strength Reliability with Clayton Copula and Generalized Rayleigh Marginals

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Abstract:

This study estimates stress-strength reliability when the stress and strength variables are dependent. The dependence is modeled by the Clayton copula, which captures lower-tail association, and the marginals follow the generalized Rayleigh distribution. A unified framework is developed that includes maximum likelihood estimation, bootstrap confidence intervals, and Bayesian estimation under different loss functions using a Metropolis-within-Gibbs sampler. A Monte Carlo simulation evaluates the performance of the estimators, and a real data example shows the practical use of the model. The findings indicate that ignoring dependence can lead to biased reliability estimates.

Keywords: Generalized Rayleigh distribution, Clayton copula, stress-strength reliability, Bayesian inference.

Mathematics Subject Classification (2010): 62N05, 62F15, 62H05.

1 Introduction

The stress–strength reliability index $R = P(X < Y)$, with X and Y denoting stress and strength, was introduced by Birnbaum [Birnbaum \(1956\)](#) and Birnbaum and McCarty [Birnbaum and McCarty \(1958\)](#). Early methods [Mazumdar \(1970\)](#) were extended to censored data [Valiollahi et al. \(2013\)](#); [Basirat et al. \(2015\)](#).

Dependence, often present in practice, is modeled via copulas [Nelsen \(2006\)](#). Recent advances address multicomponent systems [Wang and Yan \(2024\)](#); [Zhang et al. \(2025\)](#); [Akgül \(2025\)](#); [Wang and Yan \(2025\)](#); [Zhang and Yan \(2025\)](#); [Cai et al. \(2023\)](#); [Alshehri \(2026\)](#), motivating copula-based frequentist and Bayesian reliability analyses.

The Clayton copula captures lower-tail dependence, relevant in low-strength, high-stress regimes. The generalized Rayleigh (GR) distribution flexibly models skewed reliability data. No study has investigated stress–strength reliability with GR marginals under both dependent and independent structures.

We estimate R for GR marginals with Clayton copula using MLE and Bayesian inference under various loss functions with a Metropolis-within-Gibbs sampler, evaluated via simulations and real data. Contributions are: (i) novel GR–Clayton formulation with lower-tail dependence; (ii) development and comparison of MLE and Bayesian frameworks; (iii) comprehensive simulations; (iv) real-data validation.

Section 2 reviews the Clayton copula. Section 3 presents the GR distribution. Section 4 details estimation. Sections 5 and 6 provide simulations and application. Section 7 concludes.

2 Copula approach

A copula $C : [0, 1]^2 \rightarrow [0, 1]$ is a bivariate distribution with uniform marginals [Sklar \(1959\)](#); [Nelsen \(2006\)](#). By Sklar’s theorem, any bivariate distribution H with continuous marginals $F(x)$ and $G(y)$ decomposes as $H(x, y) = C(F(x), G(y))$, with joint density $h(x, y) = c(F(x), G(y))f(x)g(y)$, where $c(u, v) = \partial^2 C(u, v) / \partial u \partial v$.

We use the Clayton copula [Clayton \(1978\)](#), an Archimedean copula [Nelsen \(2006\)](#) given by

$$C(u, v; \phi) = (u^{-\phi} + v^{-\phi} - 1)^{-1/\phi}, \quad \phi \in [-1, \infty) \setminus \{0\},$$

with density

$$c(u, v) = (\phi + 1)(uv)^{-\phi-1} (u^{-\phi} + v^{-\phi} - 1)^{-(2\phi+1)/\phi}, \quad (2.1)$$

where $\phi = 2\tau/(1 - \tau)$ relates to Kendall's τ . The Clayton copula exhibits lower-tail dependence, desirable when joint low realizations influence failure, unlike Gumbel or Joe (upper-tail) and Frank (no tail dependence) [Nelsen \(2006\)](#).

3 Generalized Rayleigh Distribution

The two-parameter GR distribution [Surles and Padgett \(2001\)](#) is a special case of the generalized Weibull distribution. A random variable $X \sim GR(\alpha, \lambda)$ has pdf and cdf

$$f(x; \alpha, \lambda) = 2\alpha\lambda^2 x e^{-(\lambda x)^2} \left(1 - e^{-(\lambda x)^2}\right)^{\alpha-1}, \quad x > 0, \quad (3.1)$$

$$F(x; \alpha, \lambda) = \left(1 - e^{-(\lambda x)^2}\right)^\alpha, \quad x > 0, \quad (3.2)$$

where $\alpha, \lambda > 0$ are shape and scale parameters. The distribution is right-skewed, with mode and median decreasing in λ and increasing in α .

4 Estimation of R

4.1 Independent case

For independent $X \sim GR(\alpha_1, \lambda_1)$, $Y \sim GR(\alpha_2, \lambda_2)$, $R_I = \int_0^\infty \int_0^y f_X(x) f_Y(y) dx dy$ is evaluated numerically.

4.2 Dependent case under Clayton copula

Under the Clayton copula with parameter ϕ , $F(x) = (1 - e^{-(\lambda_1 x)^2})^{\alpha_1}$, $G(y) = (1 - e^{-(\lambda_2 y)^2})^{\alpha_2}$,

$$R_C = \int_0^\infty \int_0^y c(F(x), G(y)) f(x) g(y) dx dy,$$

where $c(u, v)$ is given by [\(2.1\)](#). Numerical integration uses adaptive Gauss–Kronrod quadrature in R (tolerances 10^{-6} , errors $< 10^{-5}$).

4.3 MLE of R_C

The log-likelihood based on $\{(x_i, y_i)\}_{i=1}^n$ is

$$\begin{aligned} L(\boldsymbol{\vartheta}) = & 2n(\ln \lambda_1 + \ln \lambda_2) + n \ln(4\alpha_1\alpha_2(\phi + 1)) + \sum_{i=1}^n (\ln x_i + \ln y_i) - \sum_{i=1}^n [(\lambda_1 x_i)^2 + (\lambda_2 y_i)^2] \\ & - (\phi\alpha_1 + 1) \sum_{i=1}^n \ln(1 - e^{-(\lambda_1 x_i)^2}) - (\phi\alpha_2 + 1) \sum_{i=1}^n \ln(1 - e^{-(\lambda_2 y_i)^2}) \\ & - \frac{2\phi + 1}{\phi} \sum_{i=1}^n \ln[(1 - e^{-(\lambda_1 x_i)^2})^{-\phi\alpha_1} + (1 - e^{-(\lambda_2 y_i)^2})^{-\phi\alpha_2} - 1], \end{aligned}$$

where $\boldsymbol{\vartheta} = (\alpha_1, \lambda_1, \alpha_2, \lambda_2, \phi)$. MLEs solve the score equations (standard differentiation; omitted).

By invariance, $\hat{R}_C = R_C(\hat{\alpha}_1, \hat{\lambda}_1, \hat{\alpha}_2, \hat{\lambda}_2, \hat{\phi})$.

4.4 Bootstrap confidence intervals

Parametric bootstrap confidence intervals for R_C are constructed by resampling from the fitted Clayton–GR model and re-estimating all parameters via MLE in each replication. Both percentile (Boot-p) and bootstrap- t (Boot- t) intervals are obtained; the latter uses a plug-in variance estimate to form Studentized intervals. Full details follow standard procedures.

4.5 Bayes estimation of R_C

With independent uniform priors on bounded intervals, posterior samples are drawn via Metropolis-within-Gibbs MCMC [Metropolis et al. \(1953\)](#). Estimates under standard loss functions are: posterior mean (SEL), posterior median (AEL), $\hat{R}_{C,LIL} = -\frac{1}{c} \ln E(e^{-cR_C} \mid \text{data})$ (LINEX), and posterior mode (MAP). Entropy loss corresponds to LINEX with $c = 1$.

5 Simulation Study

A Monte Carlo study evaluates MLE and Bayesian estimators of $R = P(X < Y)$ under the GR–Clayton model. Paired samples are generated via $X_i = F_{GR,1}^{-1}(U_i)$, $Y_i = F_{GR,2}^{-1}(V_i)$ with $(U_i, V_i) \sim C_\phi$. For each setting, 1000 replications are used.

MLE results (Table 1): MLEs exhibit small bias; SD and MSE decrease with n . $\hat{R}_C$ has negligible bias, with SD and MSE decreasing systematically.

Bayesian results (Table 2): LINEX achieves the smallest MSE; SEL outperforms AEL, ENL, and MAP; ENL yields the largest MSE.

Overall, Bayesian estimators (LINEX, SEL) achieve lower MSE in small samples; all methods converge as n increases; $n \geq 30$ provides reliable performance.

Table 1: Bias and MSE of MLEs and Bootstrap for selected settings.

n	α_1	λ_1	α_2	λ_2	ϕ	R_{true}	MLE		Boot	
							Bias	MSE	EC	AIL
For each parameter, entries are: SD / Bias / MSE.										
20	1	1	1	1	2	0.5000	0.0092	0.0103	0.9460	0.0202
	0.43/0.16/0.21	0.15/0.05/0.02	0.15/0.05/0.02	0.15/0.06/0.03	0.98/0.19/0.99					
20	5	5	1	1	5	0.7143	-0.0022	0.0005	0.9460	0.0717
	3.54/1.30/14.06	0.45/0.16/0.23	0.45/0.16/0.23	0.14/0.06/0.02	2.02/0.55/4.40					
30	1	1	1	1	2	0.5000	-0.0082	0.0058	0.9660	0.0070
	0.30/0.11/0.10	0.11/0.04/0.01	0.11/0.04/0.01	0.12/0.03/0.02	0.74/0.16/0.58					
30	5	5	1	1	5	0.7143	-0.0018	0.0004	0.9520	0.0640
	2.04/0.84/4.95	0.36/0.12/0.14	0.36/0.12/0.14	0.12/0.04/0.02	1.57/0.23/2.52					
40	1	1	1	2	2	0.5000	0.0012	0.0005	0.9620	0.0836
	0.25/0.08/0.07	0.11/0.03/0.01	0.24/0.07/0.06	0.19/0.05/0.04	0.65/0.10/0.44					
40	5	5	1	2	5	0.7143	-0.0056	0.0026	0.9560	0.0879
	1.62/0.58/2.97	0.33/0.10/0.12	0.24/0.07/0.06	0.21/0.06/0.05	1.33/0.28/1.85					
50	1	1	1	2	2	0.5000	-0.0007	0.0004	0.9650	0.0062
	0.20/0.06/0.05	0.09/0.02/0.01	0.20/0.07/0.04	0.18/0.05/0.04	0.59/0.04/0.35					
50	5	5	1	2	5	0.7143	-0.0038	0.0021	0.9400	0.1005
	1.34/0.44/1.98	0.27/0.07/0.08	0.22/0.06/0.05	0.19/0.06/0.04	1.18/0.23/1.45					

Table 2: Bias and MSE of Bayes estimators under different loss functions.

n	α_1	λ_1	α_2	λ_2	ϕ	Bias					MSE				
						Abs	Ent	MAP	Sq	Lin	Abs	Ent	MAP	Sq	Lin
20	1	1	1	1	1	.061	1.700	.248	.050	-.015	.004	2.889	.061	.003	.016
20	1	1	1	1	2	.061	1.700	.248	.050	-.015	.004	2.889	.067	.003	.000
20	1	1	1	1	3	.085	1.686	.268	.069	.003	.007	2.843	.082	.007	.000
20	3	3	5	1	3	.001	0.310	.018	.001	.000	.000	0.125	.003	.000	.000
20	5	5	1	1	5	.090	1.632	.145	.072	.004	.009	2.664	.019	.007	.000
20	3	1	1	1	2	.000	5.197	.019	.000	.000	.000	37.012	.000	.000	.000
20	1	3	1	1	2	.129	1.293	.361	.110	-.057	.017	1.672	.139	.015	.003
20	1	1	3	1	2	.000	5.900	.017	.000	.000	.000	34.814	.005	.000	.000
20	3	3	5	1	2	.000	0.140	.136	.000	.000	.000	0.024	.005	.000	.000
40	1	1	1	1	1	-.056	1.475	.144	-.058	-.130	.003	2.177	.021	.003	.000
40	1	1	1	1	2	.059	1.608	.259	.058	-.013	.004	2.587	.061	.003	.000
40	1	1	1	1	3	.082	1.545	.286	.080	.006	.007	2.386	.079	.005	.000
40	3	3	5	2	3	.001	0.420	.017	.001	.000	.000	0.185	.002	.000	.000
40	5	5	1	2	5	.094	1.457	.125	.086	.008	.008	2.123	.014	.005	.000
40	3	1	1	1	2	.000	6.073	.017	.000	.000	.000	27.884	.000	.000	.000
40	1	3	1	1	2	.125	1.248	.373	.122	-.048	.016	1.558	.120	.014	.002
40	1	1	3	1	2	.134	1.196	.383	.132	-.056	.000	1.431	.002	.000	.000
40	3	3	5	2	2	.000	0.190	.125	.000	.000	.000	0.038	.003	.000	.000

6 Application to Real Data

We use UEFA Champions League data (2004–2006) from Meintanis [Meintanis \(2007\)](#), where X is the time (minutes) of the first goal from a direct kick and Y is the home team's first goal time ($R = P(X < Y)$). The data are:

X : 26, 63, 19, 66, 40, 49, 8, 69, 39, 82, 72, 66, 25, 41, 16, 18, 22, 42, 36, 34, 53, 54, 51, 76, 64, 26, 16, 44, 25, 55, 49, 24, 44, 42, 27, 28, 2.

Y : 20, 18, 19, 85, 40, 49, 8, 71, 39, 48, 72, 62, 9, 3, 75, 18, 14, 42, 52, 34, 39, 28, 64, 15, 48, 16, 13, 14, 11, 49, 24, 30, 3, 47, 28, 2.

The Pearson correlation is 0.469 ($p = 0.0034$). Both marginals fit the GR distribution (K-S $p > 0.92$) and the Clayton copula parameter is $\hat{\phi} = 0.76$ ($p \leq 0.01$; [Table 3](#)).

Table 3: Fitting real data to the GR and Clayton copula.

Marginal	K-S	p -value	$\hat{\alpha}$	$\hat{\lambda}$						
					Copula	$\hat{\phi}$	p -value	logLik	AIC	BIC
X (stress)	0.0871	0.9414	1.0117	0.0220						
Y (strength)	0.0894	0.9285	0.5721	0.0207	Clayton	0.76	≤ 0.01	3.23	-4.46	-2.85

Table 4: Estimations of R_C for the UEFA Champions League data.

Method	R_C
MLE	0.3631
Bayesian: MAP	0.3675
Bayesian: Linex	0.3734
Bayesian: Absolute error	0.3749
Bayesian: Entropy	0.3756
Bayesian: Squared error	0.3778

7 Conclusion

This study proposed estimation of $R = P(X < Y)$ under Clayton copula dependence with GR marginals, using MLE, bootstrap (Boot-p, Boot-t), and Bayesian methods under various loss functions. Simulations showed bootstrap and Bayesian methods, particularly LINEX, outperform classical approaches in small samples. Real UEFA Champions League data validated the methodology.

The framework generalizes directly to other Archimedean copulas (Gumbel, Frank) by substituting the copula density in the likelihood, bootstrap, and MCMC procedures, and can be adapted

to elliptical copulas (Gaussian, t) with appropriate numerical integration, demonstrating broad applicability across reliability, engineering, and applied statistics.

Future work may explore alternative copulas, flexible marginals, and applications in biostatistics or financial risk modeling.

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